

# Vineeth Appikarla

AI/ML Engineer

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## PROFESSIONAL SUMMARY

Experienced AI/ML Engineer with 4+ years of experience designing, developing, and deploying end-to-end machine learning solutions for financial services and enterprise applications. Skilled in building scalable ML pipelines, feature engineering, predictive modeling, anomaly detection, and explainable AI, leveraging Python, SQL, TensorFlow, PyTorch, XGBoost, and cloud platforms. Adept at collaborating with cross-functional teams, translating business requirements into data-driven solutions, and optimizing operational efficiency through automated ML workflows. Proven ability to deliver high-impact results in AI/ML projects, improving decision accuracy, reducing manual effort, and enhancing model performance across complex datasets.

## TECHNICAL SKILLS

**Programming Languages:** Python (NumPy, Pandas, Regex, Pickle, OS, JSON), SQL (PostgreSQL, MySQL, SQLite), Bash

**Machine Learning Algorithms:** Linear Regression, Ridge/Lasso/Polynomial Regression, Logistic Regression (Binary & Multiclass), Decision Tree, Random Forest, SVM, KNN, Naive Bayes, GBM, XGBoost, LightGBM, CatBoost

**ML Libraries & Frameworks:** Scikit-learn, TensorFlow, Keras, PyTorch, Statsmodels, XGBoost, LightGBM, CatBoost, AWS Neuron

**NLP & Text Analytics:** SpaCy, NLTK, Gensim, TF-IDF, Bag-of-Words, Cosine Similarity, TextBlob, BERT, LayoutLM

**Data Handling & Databases:** ETL, Data Cleaning, Feature Engineering, EDA, Aggregations, Joins, Data Transformation, Pandas, NumPy, MongoDB, PostgreSQL, SQLite, Amazon Redshift, Firebase, InfluxDB

**Model Deployment & MLOps:** AWS SageMaker, Docker, Kubernetes, Apache Airflow, Flask, FastAPI, MLflow, GitHub Actions, Jenkins, CI/CD Pipelines

**Cloud & DevOps:** AWS (EC2, S3, Lambda, SageMaker, RDS, IAM), Git, CI/CD Integration, AWS CloudWatch

**Visualization & Reporting:** Tableau, Power BI, Kibana, Matplotlib, Seaborn, SHAP, Advanced Excel

**Collaboration & Tools:** JIRA, Confluence, TestRail, Microsoft Teams, Microsoft Visual Studio, Jupyter Notebooks, PyCharm

## PROFESSIONAL EXPERIENCE

### AI/ML Engineer | Northern Trust, USA

Mar 2025 - Present

- Developed and deployed an end-to-end AI/ML platform integrating portfolio risk detection and intelligent asset rebalancing to optimize investment decisions; leveraged Python, TensorFlow, Scikit-learn, XGBoost, and AWS to streamline model workflows, resulting in a 32% improvement in anomaly detection accuracy and 27% faster portfolio optimization cycles across institutional and private banking divisions.
- Engineered and operationalized scalable data pipelines using Kafka, Airflow, and AWS Redshift to consolidate multi-source financial datasets, ensuring consistent ingestion of market data, client portfolios, and access logs, which reduced manual data discrepancies by 35% and enhanced data reliability for modeling and compliance reporting.
- Designed and refined ML models including autoencoders and gradient boosting frameworks for fraud risk scoring and asset return forecasting; applied SHAP-based interpretability to maintain regulatory transparency and elevated model trust and adoption among risk management teams by 40%.
- Collaborated cross-functionally with wealth managers, compliance officers, and data governance teams to translate business rules and market risk indicators into actionable ML features and optimization constraints; this synergy increased operational decision accuracy by 45%, aligning predictive outcomes with investment mandates and risk appetites.
- Implemented continuous retraining and monitoring pipelines through MLflow and Prefect, automating feedback integration from alerts and rebalancing signals; this iterative improvement framework boosted system stability by 30% and ensured adaptive learning in volatile market conditions while maintaining adherence to financial compliance standards (SOX, Basel III).

### Machine Learning Engineer | Happiest Minds Technologies, India

Sep 2021 - Aug 2023

- Architected and deployed end-to-end ML pipelines for fraud detection and credit risk modeling, integrating streaming financial data from multiple banking systems using Kafka and Spark; this initiative enhanced fraud detection precision by 38% and accelerated credit decision cycles by 42%, improving operational resilience across partner banks.
- Formulated and optimized feature engineering strategies for dynamic transaction velocity, behavioral risk scoring, and cash-flow stability, leveraging AWS S3, Redshift, and SQL-based analytics to elevate data accuracy by 36% and minimize false-positive fraud alerts by 33%, directly impacting financial trust and compliance adherence.
- Engineered scalable ML models using gradient boosting and neural architectures for fraud risk and credit scoring, deploying via Docker and Kubernetes for real-time inference; applied SHAP for interpretability to ensure transparency with regulatory teams and increased explainability adoption by 40% within risk operations.
- Collaborated with financial analysts and compliance units to translate AML/KYC regulations and credit policy constraints into algorithmic business rules, aligning model thresholds and audit trails with SOX, Basel III, and GDPR frameworks to strengthen governance and reduce compliance escalations.
- Implemented continuous monitoring and retraining pipelines with Airflow and MLflow to capture model drift, feedback labels, and system anomalies, establishing a robust feedback ecosystem that improved long-term model stability by 31% and sustained data governance maturity across financial workflows.
- Spearheaded the integration of model outputs into fraud case management dashboards using Power BI and Elasticsearch, empowering investigators and loan officers with real-time insights and interactive visualizations that cut manual investigation time by 45% and boosted approval efficiency for near-prime customers by 28%.

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| <b>Jr. Machine Learning Engineer   Newgen Software, India</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Jun 2019 - Aug 2021</b> |
| <ul style="list-style-type: none"><li>• Architected and deployed scalable ML workflows for OmniDocs-Recon and LoanSense-AI to automate invoice reconciliation and credit risk scoring, utilizing OCR, NER, and predictive analytics to transform unstructured financial data into structured insights, which improved reconciliation accuracy by 78% and accelerated loan approvals by 62%.</li><li>• Synthesized and engineered financial data pipelines integrating invoice, payment, and credit records using Python, SQL, and Apache Kafka; enhanced data integrity across ERP and banking systems, enabling reliable end-to-end automation within ECM and BPM workflows for treasury and lending operations.</li><li>• Devised and refined ML models using LightGBM and Transformer-based architectures for document classification, supplier normalization, and credit risk evaluation; implemented explainable AI techniques such as SHAP to enhance transparency and reduced manual validation efforts by 55% while boosting stakeholder trust in model-driven decisions.</li><li>• Operationalized and monitored production-grade ML models through Airflow, MLflow, Docker, and Kubernetes, automating retraining cycles and anomaly detection within large-scale invoice and loan datasets to ensure consistent system performance and regulatory compliance under SOX and Basel III standards.</li><li>• Collaborated cross-functionally with financial analysts, risk managers, and IT teams to align model outputs with AML/KYC policies, credit scoring parameters, and audit traceability, integrating the workflows with Newgen’s OmniFlow for automated posting and approval of reconciliations and credit decisions.</li><li>• Consolidated data insights through Power BI and Kibana dashboards, presenting reconciliation rates, fraud alerts, and portfolio risk metrics in real time, which optimized exception resolution efficiency by 70% and enhanced financial governance visibility across multi-entity banking operations.</li></ul> |                            |

EDUCATION

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| <b>Masters of Science in Data Science</b><br>DePaul University, Chicago, IL, USA                                                           | <b>Sep 2023 - Jun 2025</b> |
| <b>Bachelors of Technology in Information Technology</b><br>Anil Neerukonda Institute of Technology and Sciences, Visakhapatnam, AP, India | <b>Jun 2017 - Jun 2021</b> |